

COMEBACK MOMENTUM, CHART RE-ENTRY, AND FANDOM INTENSITY ANALYSIS OF SOUTH KOREA TOP 50 PLAYLIST

Abstract

The South Korean music industry, particularly the K-Pop sector, has developed a unique streaming ecosystem driven by highly engaged fan communities, frequent artist comebacks, and recurring promotional cycles. Unlike traditional music markets where chart success is primarily determined by sustained popularity, the South Korean market exhibits repeated momentum bursts resulting from fandom-driven engagement. Songs frequently exit and re-enter charts, making chart longevity alone an insufficient measure of success.

This study analyzes the South Korea Top 50 Playlist dataset to investigate chart re-entry behavior, comeback momentum, and fandom intensity. The dataset contains daily playlist rankings, popularity metrics, album characteristics, and content attributes. A framework consisting of six key performance indicators (KPIs) was developed, including Re-Entry Frequency, Momentum Spike Score, Post-Comeback Retention Days, Rank Recovery Speed, Album Comeback Advantage Index, and Fandom Intensity Proxy Score.

The analysis reveals that several songs repeatedly regain chart positions after temporary exits, indicating strong audience loyalty and organized fan engagement. The findings demonstrate the importance of fandom-driven consumption patterns and provide strategic recommendations for Atlantic Recording Corporation to optimize release planning, promotional campaigns, and artist positioning within the South Korean market.

Keywords: K-Pop, Music Analytics, Chart Re-Entry, Streaming Analysis, Fandom Engagement, Comeback Momentum

Chapter 1: Introduction

1.1 Background

The digital transformation of the music industry has fundamentally changed the way audiences consume music. Streaming platforms have become the dominant mode of music distribution, providing artists and record labels with access to large-scale listener behavior data. While many global music markets exhibit consumption patterns driven by organic discovery and recommendation algorithms, South Korea presents a distinct environment characterized by highly organized fandom participation.

K-Pop fandoms actively engage in streaming campaigns, promotional activities, social media engagement, and coordinated support initiatives designed to maximize chart performance. As a result, songs frequently experience multiple popularity cycles rather than following a traditional pattern of gradual growth and decline.

The South Korean music ecosystem therefore requires specialized analytical frameworks capable of measuring comeback momentum, re-entry behavior, and fandom engagement intensity.

1.2 Industry Context

Atlantic Recording Corporation operates within an increasingly competitive global music landscape. Understanding region-specific audience behavior has become essential for effective release planning and marketing strategy. Traditional performance indicators such as total streams and chart longevity often fail to capture the dynamic nature of fandom-driven engagement observed within South Korea.

Songs may temporarily disappear from major playlists before returning due to promotional events, award ceremonies, anniversaries, live performances, or organized fan campaigns. These re-entry events represent valuable signals of audience commitment and long-term popularity.

1.3 Research Objectives

The primary objectives of this study are:

1. To identify songs exhibiting repeated chart re-entry behavior.
2. To measure comeback momentum following chart re-entry.
3. To evaluate sustainability of popularity surges.
4. To compare comeback performance between singles and album tracks.
5. To investigate the effect of explicit content on chart performance.
6. To estimate fandom engagement intensity using playlist data.

1.4 Research Questions

The study seeks to answer the following questions:

- Which songs demonstrate the highest chart re-entry frequency?
- How strong are momentum spikes following chart re-entry?
- How long do comeback effects remain active?
- Do singles outperform album tracks during comeback periods?
- Does explicit content influence chart resilience?
- Can fandom intensity be estimated through chart behavior?

Chapter 2: Literature Review

2.1 Music Streaming Analytics

Music streaming analytics has become an important area of research due to the availability of large-scale listener behavior data. Previous studies have examined chart prediction, popularity forecasting, recommendation systems, and listener retention using streaming metrics.

Researchers have found that streaming data can reveal valuable insights into audience preferences, consumption patterns, and content performance. However, many analytical frameworks remain focused on cumulative popularity rather than recurring engagement cycles.

2.2 Fandom-Driven Consumption

Fandom participation plays a significant role within the K-Pop industry. Unlike passive music consumption, K-Pop fandoms frequently engage in coordinated streaming activities intended to improve chart performance and increase visibility for artists.

These behaviors create unique chart dynamics in which songs can repeatedly gain popularity despite periods of inactivity.

2.3 Chart Re-Entry Behavior

Chart re-entry refers to the process by which a song exits a ranking system and subsequently returns at a later date. Existing literature suggests that re-entry events may be influenced by promotional campaigns, viral social media activity, award recognition, live performances, and fandom mobilization.

However, relatively limited research has examined re-entry behavior within South Korea's streaming ecosystem.

2.4 Research Gap

Most existing studies prioritize popularity accumulation and chart longevity. Few frameworks directly measure comeback momentum or fandom intensity using chart re-entry patterns. This study addresses this gap by introducing a momentum-centric analytical approach specifically designed for the South Korean market.

Chapter 3: Problem Statement and Dataset Description

3.1 Problem Statement

Despite access to daily playlist rankings, Atlantic Recording Corporation lacks visibility into:

- Re-entry frequency patterns
- Comeback momentum intensity
- Sustainability of chart recoveries
- Fandom-driven engagement behavior
- Content attribute impact on chart performance

Without these insights, strategic decisions may not fully align with the characteristics of the South Korean music market.

3.2 Dataset Description

The South Korea Top 50 Playlist dataset contains daily playlist snapshots and includes the following variables:

- Date
- Playlist Position
- Song Title
- Artist
- Popularity Score
- Duration (milliseconds)
- Album Type
- Number of Tracks

- Explicit Content Indicator
- Album Cover URL

The dataset consists of approximately 27,800 observations and more than 500 unique songs.

3.3 Data Preparation

Data preprocessing involved:

- Date standardization
- Song–artist normalization
- Duration conversion to minutes
- Validation of popularity scores
- Construction of unique song identifiers

These steps ensured consistency throughout the analytical process.

Chapter 4: Research Methodology

4.1 KPI 1: Re-Entry Frequency

Re-Entry Frequency measures the number of times a song returns to the chart after disappearing for at least one day.

Re-Entry Frequency = Number of Returns After Disappearance

4.2 KPI 2: Momentum Spike Score

Momentum Spike Score evaluates the strength of a comeback using rank improvement and popularity growth.

Momentum Score = $0.6 \times \text{Rank Jump} + 0.4 \times \text{Popularity Increase}$

4.3 KPI 3: Post-Comeback Retention Days

Measures the number of days a song remains on the chart after re-entry.

4.4 KPI 4: Rank Recovery Speed

Measures the number of days required to achieve peak chart position following re-entry.

4.5 KPI 5: Album Comeback Advantage Index

Compares average comeback performance of album tracks and singles.

4.6 KPI 6: Fandom Intensity Proxy Score

Fandom Intensity Score combines re-entry frequency, momentum strength, retention performance, and recovery efficiency into a single engagement indicator.

Chapter 5: Results and Analysis

5.1 Overview of Findings

The analysis focused on identifying chart re-entry behavior, comeback momentum patterns, fandom-driven engagement, and content-related performance differences within the South Korea Top 50 Playlist dataset.

The dataset revealed that chart success within South Korea is not solely determined by continuous popularity. Instead, many songs experienced multiple cycles of disappearance and resurgence, indicating recurring audience engagement and fandom participation.

The results support the hypothesis that fandom-driven consumption plays a significant role in shaping chart performance within the South Korean music ecosystem.

5.2 Re-Entry Frequency Analysis

Re-entry frequency was calculated by identifying songs that disappeared from the playlist and later returned.

The analysis revealed several songs exhibiting exceptionally high re-entry frequencies.

Top 10 Songs by Re-Entry Frequency

Rank	Song	Re-Entries
1	Our Blues, Our Life – Lim Young Woong	31
2	Ditto – NewJeans	27
3	A bientot – Lim Young Woong	24
4	New Jeans – NewJeans	23
5	Hype Boy – NewJeans	21
6	Interlude : Dive – Jimin	21
7	Rainbow – Lim Young Woong	19
8	ETA – NewJeans	18
9	Cruel Summer – Taylor Swift	16
10	Somebody – Jung Kook	15

Interpretation

The presence of NewJeans and Lim Young Woong across multiple positions demonstrates the strength of their fan communities. Songs associated with these artists repeatedly regained chart positions despite periods of inactivity. This behavior indicates strong audience loyalty and long-term engagement.

5.3 Comeback Momentum Analysis

Momentum analysis evaluated the ability of songs to regain popularity after re-entry.

Songs exhibiting high re-entry frequencies generally demonstrated stronger comeback momentum compared with songs that appeared only once.

The results suggest that fandom mobilization, promotional activities, and recurring listener engagement contribute significantly to chart recovery performance.

Artists with established fan communities demonstrated faster and stronger chart recoveries following re-entry events.

5.4 Retention and Sustainability Analysis

Retention analysis examined how long songs remained within the Top 50 after returning to the chart.

Songs exhibiting higher re-entry frequencies also tended to remain charted for longer periods following their return.

This observation indicates that successful re-entry events often generate sustained engagement rather than temporary popularity spikes.

The relationship between re-entry frequency and chart persistence suggests that audience commitment plays an important role in maintaining visibility within the playlist.

5.5 Single vs Album Analysis

The dataset was analyzed to compare comeback behavior between singles and album tracks.

Findings

- Singles demonstrated stronger chart recovery behavior.
- Singles appeared to benefit more from concentrated promotional campaigns.
- Album tracks generally exhibited lower comeback intensity.

Interpretation

The findings suggest that title tracks and promoted singles receive greater fandom attention than secondary album tracks.

This aligns with industry practices where marketing resources are concentrated on lead singles.

5.6 Explicit vs Non-Explicit Analysis

The dataset included both explicit and non-explicit songs.

Findings

Non-explicit songs demonstrated stronger chart resilience and re-entry behavior compared with explicit content.

Interpretation

The South Korean market appears to favor broadly accessible content that appeals to multiple demographic groups.

Unlike several Western streaming markets where explicit content frequently dominates chart performance, South Korea exhibits stronger engagement with clean content.

5.7 Fandom Intensity Analysis

Fandom intensity was estimated using chart re-entry behavior, comeback persistence, and engagement stability.

Key Observations

Artists consistently appearing across re-entry rankings exhibited stronger fandom intensity indicators.

The analysis suggests that fandom activity significantly influences chart resilience and comeback success.

High Fandom Intensity Artists

- NewJeans
- Lim Young Woong
- Jimin
- Jung Kook

These artists demonstrated repeated chart recoveries and sustained audience engagement throughout the observation period.

Chapter 6: Discussion

The findings reveal that South Korea's streaming ecosystem differs substantially from traditional chart systems.

Unlike conventional popularity models where chart success gradually declines over time, many songs in the dataset exhibited recurring cycles of growth and recovery.

This behavior highlights the importance of fandom-driven engagement as a core driver of chart performance.

The repeated success of artists such as NewJeans and Lim Young Woong suggests that strong fan communities can sustain audience attention long after initial release periods.

The results further demonstrate that chart re-entry behavior provides valuable information beyond traditional popularity metrics. Songs capable of returning to the chart multiple times possess characteristics associated with enduring audience interest.

Additionally, the superior performance of singles indicates that focused promotional campaigns remain highly effective within the South Korean market.

The explicit content analysis further suggests that cultural and demographic preferences influence chart outcomes.

Overall, the findings support the argument that chart performance should be evaluated using momentum-based indicators rather than solely relying on popularity rankings.

Chapter 7: Recommendations for Atlantic Recording Corporation

Based on the findings of this study, the following recommendations are proposed.

7.1 Monitor Re-Entry Behavior

Traditional popularity tracking should be expanded to include chart re-entry analysis.

Songs demonstrating repeated re-entry behavior should be prioritized for continued promotion and audience engagement strategies.

7.2 Develop Fandom-Centric Marketing Campaigns

Fandom participation significantly influences chart performance.

Marketing initiatives should focus on fan engagement through:

- Digital campaigns
- Social media interaction
- Exclusive content
- Community-building activities

7.3 Optimize Release Scheduling

Releases should be strategically aligned with:

- Award ceremonies
- Live performances
- Anniversary events
- Promotional appearances

These events can increase the likelihood of chart resurgence.

7.4 Focus Resources on High-Potential Singles

Singles demonstrated stronger comeback characteristics than album tracks.

Marketing investments should prioritize songs with the highest potential for sustained audience engagement.

7.5 Implement Momentum-Based KPIs

Atlantic Recording Corporation should incorporate:

- Re-Entry Frequency
- Momentum Spike Score
- Retention Days
- Fandom Intensity Score

into future artist performance evaluation frameworks.

Chapter 8: Conclusion

This study examined chart re-entry behavior, comeback momentum, and fandom intensity within the South Korea Top 50 Playlist.

The findings demonstrate that chart performance within South Korea extends beyond traditional popularity measures and is strongly influenced by recurring engagement cycles.

The analysis identified multiple songs exhibiting repeated re-entry behavior, suggesting that audience loyalty and fandom participation contribute significantly to long-term chart resilience.

Artists such as NewJeans, Lim Young Woong, Jimin, and Jung Kook consistently demonstrated strong engagement indicators across multiple analytical dimensions.

The study further revealed that singles generally outperform album tracks during comeback periods and that non-explicit content exhibits stronger chart persistence than explicit content.

The proposed KPI framework provides a practical approach for measuring comeback effectiveness and fandom engagement within highly competitive streaming environments.

The findings can support Atlantic Recording Corporation in developing data-driven release strategies tailored specifically to the characteristics of the South Korean market.

Executive Summary

The objective of this project was to analyze comeback momentum, chart re-entry behavior, and fandom intensity within the South Korea Top 50 Playlist.

A dataset containing playlist rankings, popularity scores, album characteristics, and content attributes was analyzed to identify recurring engagement patterns.

The study introduced six key performance indicators:

1. Re-Entry Frequency

2. Momentum Spike Score
3. Post-Comeback Retention Days
4. Rank Recovery Speed
5. Album Comeback Advantage Index
6. Fandom Intensity Proxy Score

The results revealed that chart performance in South Korea is strongly influenced by fandom-driven engagement. Several songs repeatedly re-entered the chart after periods of absence, indicating recurring audience activation.

NewJeans and Lim Young Woong emerged as the strongest performers in terms of chart resilience and re-entry behavior.

The findings provide actionable insights that can assist Atlantic Recording Corporation in optimizing artist promotion strategies, release planning, and audience engagement initiatives.

Future Scope

Future studies may incorporate:

- Spotify streaming counts
- YouTube view statistics
- TikTok engagement metrics
- Social media sentiment analysis
- Concert and tour schedules
- Award show appearances

Integrating these factors may improve the accuracy of fandom intensity estimation and comeback prediction models.

References

1. IFPI Global Music Report.
2. Spotify Streaming Analytics Documentation.
3. Billboard Chart Methodology.
4. Research studies on K-Pop fandom engagement.
5. Music streaming behavior and recommendation system literature.
6. South Korean digital music market reports.
7. Academic studies on consumer engagement in online music platforms.